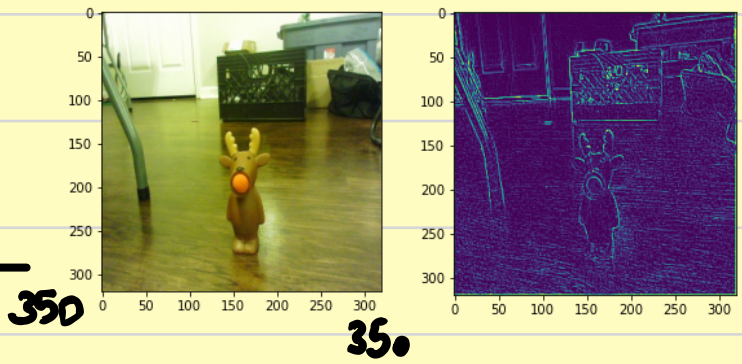
DEEP LEARNING by HAND (2)

CONVOLUTION

INPUT → FILTER → EDGES



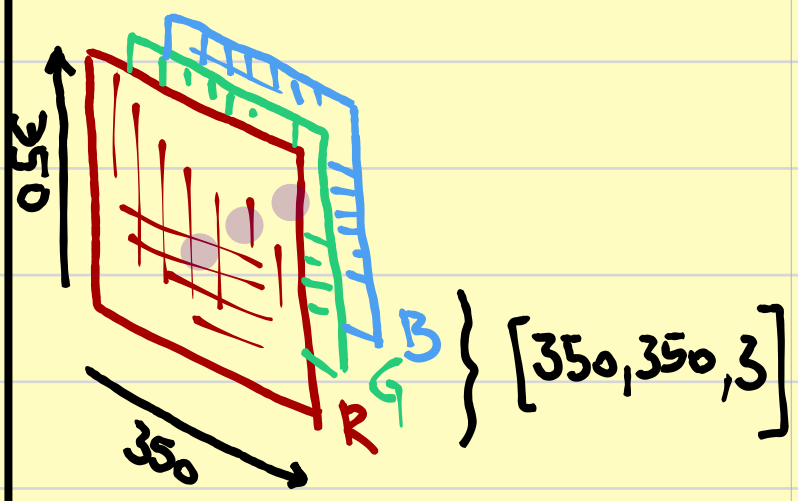
350 × 350

CONVOLUTION is a technique used in image processing. Is the process of summing each element of the input image multiplied by a filter with its neighbours.

This implies that we can measure distances btw. elements of the domain: domain must be EUCLYDEAN.

The filters are called KERNELS. In this case, a Kernel is a matrix/tensor that shapes the convolution.

WHAT IS A COLOR IMAGE?



WHAT ARE PIXELS?

1 Pixel 

The values of each channel are between [0,255] = 256.

Each pixel has 3 values superimposed. [137, 38, 205] (R G B)

KERNELS. Kernels may have many shapes and sizes. The kernels make up filters, which are a parameter in the convolutional layer (Deep learning). The kernels may be used for different purposes: . EDGE RECOGNITION / ATTENTION / BLURRING / ...

Examples (3×3):

1) EDGE — Only edges are non zero.

1	0	-1
0	0	0
-1	0	1

2) ATTENTION — The middle is highlighted.

0	-1	0
-1	4	-1
0	-1	0

Example of convolution:

R

1	1	0	2	1
1	2	1	1	0
0	1	2	1	1
1	2	2	1	1
1	0	0	1	0

Attention Kernel (3x3)

0	-1	0
-1	4	-1
0	-1	0

Convolution parameter STRIDE $S=1$.
This means we move our kernel one step ($s=1$) at a time

$$\begin{bmatrix}1&1&0\\1&2&1\\0&1&2\end{bmatrix} \cdot \begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix} \equiv \begin{matrix}1.0+1.(-1)+0.0+\\1.(-1)+2.4+1.(-1)+\\0.0+1.(-1)+2.0\end{matrix} = 4$$

$$\begin{bmatrix}1&0&2\\2&1&1\\1&2&1\end{bmatrix} \cdot \begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix} \equiv \begin{matrix}1.0+0.(-1)+2.0+\\2.(-1)+1.4+1.(-1)+\\1.0+2.(-1)+1.0\end{matrix} = -1$$

$$\begin{bmatrix}0&2&1\\1&1&0\\2&1&1\end{bmatrix} \cdot \begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix} \equiv \begin{matrix}0.0+2.(-1)+1.0+\\1.(-1)+1.4+0.(-1)+\\2.0+1.(-1)+1.0\end{matrix} = 0$$

$$\begin{bmatrix}1&2&1\\0&1&2\\1&2&2\end{bmatrix} \cdot \begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix} \equiv \ldots = -2$$

CONVOLUTED IMAGE

4	-1	0
-2	3	-1
4	3	-1

[3x3]

We have convolved a 5x5 red channel into a 3x3 red channel. With a kernel 3x3 $s=1$.

Excercise for home: same kernel and red channel but now $s=2$.

SOLUTION:

4	0
4	-1

[2x2]



IF WE INCREASE THE STRIDE WE GET LESS RESSOLUTION.

PADDING.

Padding means adding zeros all around the image...

1	1	0	2	1
1	2	1	1	0
0	1	2	1	1
1	2	2	1	1
1	0	0	1	0

...to make sure we do not loose information with convolution.

$p = 1$

Exercise for home: same kernel and red channel as above. $s = 3$, $p = 1$.

Solution:

2	6
1	-1

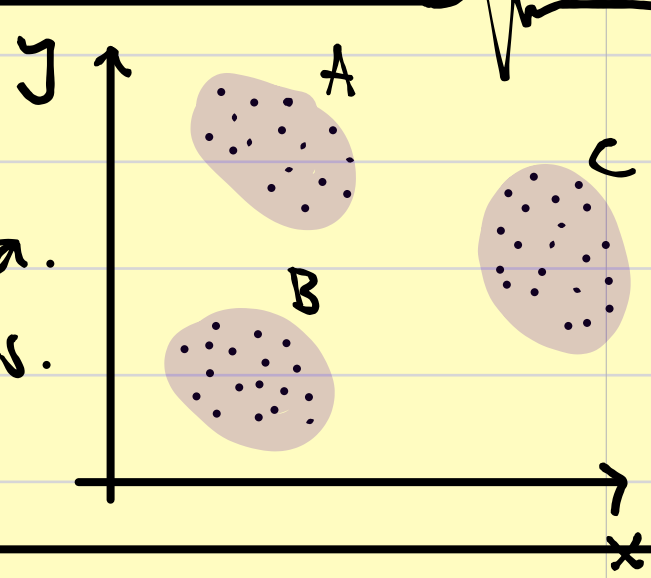
$[2, 2]$

K-MEANS CLUSTERING

Clustering algorythm builds groups of data.

k is a parameter giving the number of groups.

k needs to be defined by the user.



kMC: Step 0. Decide the number of clusters „k".

→ Step 1. Distribute the datapoints in „k" groups.

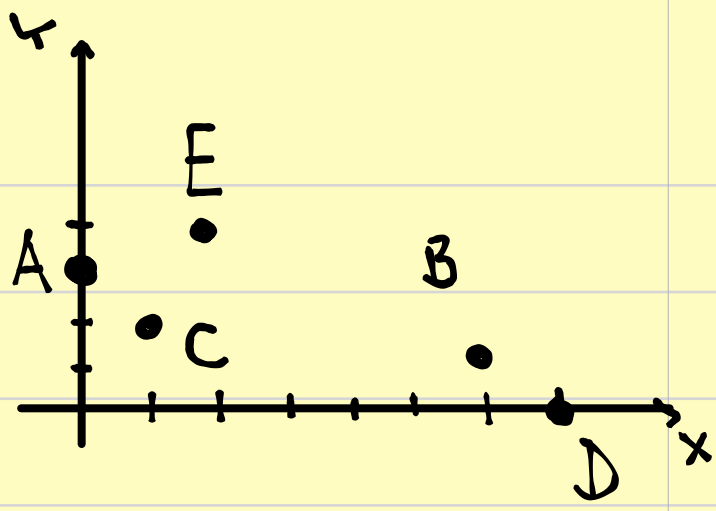
Step 2. Find the centroid of each group.

Step 3. Calculate the distance of each point to the centroids

Step 4. Cluster the points by the distances to centroids.

Step 5. Repeat to Step 1 until the groups are constant.

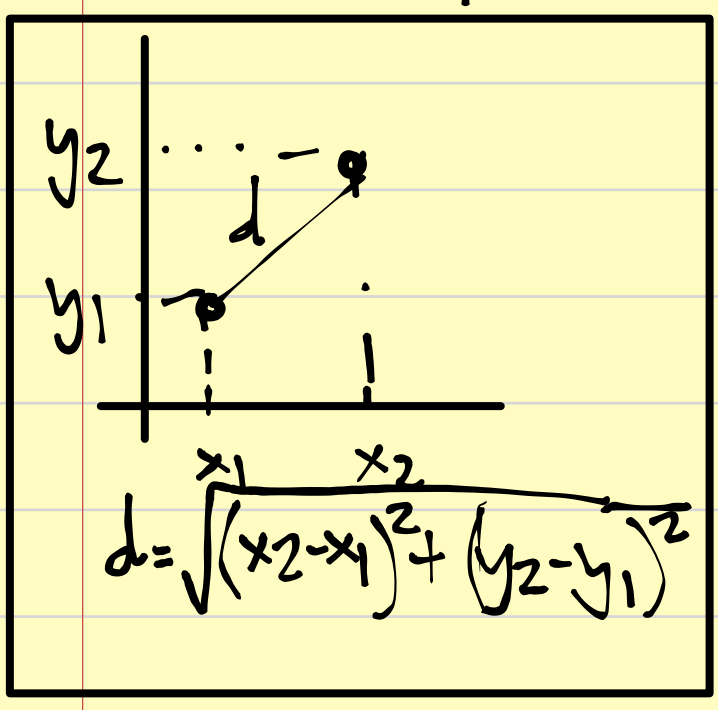
Example: A[0,3] B[6,1]
C[1,2] D[7,0]
Place 2 Warehouses E[2,4] with KMC to deliver the customers.



S0 ≡ K=2 S1 ≡ $G_1=\{ABC\}$ $G_2=\{DE\}$

S2 ≡ centroids.

$$Z_1=\left\{\frac{0+6+1}{3}, \frac{3+1+2}{3}\right\}=\{2'33, 2\} \qquad Z_2=\left\{\frac{7+2}{2}, \frac{0+4}{2}\right\}=\{4'5, 2\}$$



S3 ≡ distances

$d_{AZ1}=\sqrt{(0-2'33)^2+(3-2)^2}=2'53$; $d_{AZ2}=\sqrt{(0-4'5)^2+(3-2)^2}=4'61$

$d_{BZ1}=3'804$; $d_{BZ2}=1'803$

$d_{CZ1}=1'33$; $d_{CZ2}=4'5$

$d_{DZ1}=5'08$; $d_{DZ2}=3'201$

$d_{EZ1}=2'027$; $d_{EZ2}=3'201$

S4. new clusters! $G_1^*=\{A,C,E\}$ $G_2^*=\{B,D\}$

S2. centroids. $Z_1^*=\left\{\frac{0+1+2}{3}, \frac{3+2+4}{3}\right\}=\{1,3\}$ $Z_2^*=\left\{\frac{6+7}{2}, \frac{1+0}{2}\right\}=\{6'5, 0'5\}$

S3. distances $d_{AZ1}^*=1$ $d_{AZ2}^*=6'9$; $d_{DZ1}^*=6'71$ $d_{DZ2}^*=0'7$

$d_{BZ1}^*=5'4$ $d_{BZ2}^*=0'7$; $d_{EZ1}^*=0'7$ $d_{EZ2}^*=5'7$

$d_{CZ1}^*=1$ $d_{CZ2}^*=5'7$;

clusters remain constant: $G_1^*\{ACE\}$ $G_2^*\{BD\}$

Warehouses $z_1^* = \{1,3\}$ $z_2^* = \{6'5, 0'5\}$

4. **Ex (20 Points):** Calculate the position of 2 warehouses for the plants with the X, Y positions using known clustering algorithms:

$$[X, Y] = \{\text{Plant 1 } (2,3), \text{Plant 2 } (0,2), \text{Plant 3 } (6,2), \text{Plant 4 } (6,0)\}$$

. Please start your calculation with clusters C_1 $[Plant1, Plant3]$, and C_2 $[Plant2, Plant4]$.

Exam Question for home. SS24

$P_1[2,3]$ $k=2$ $z_1 = \left[\frac{2+6}{2}, \frac{3+2}{2}\right] = [4, 2'5]$ $z_2 = \left[\frac{6+0}{2}, \frac{0+2}{2}\right] = [3,1]$

$P_2[0,2]$

$P_3[6,2]$

$P_4[6,0]$

$d_{P_1 z_1} = 2'06$ $d_{P_1 z_2} = 2'23$

$d_{P_2 z_1} = 4'03$ $d_{P_2 z_2} = 2'23$

$d_{P_3 z_1} = 2'06$ $d_{P_3 z_2} = 3'16$

$d_{P_4 z_1} = 3'2$ $d_{P_4 z_2} = 3'16$

$G_1^* = [P_1, P_3]$

$G_2^* = [P_2 P_4]$

Groups remain constant ✓

Warehouses are placed $z_1 = [4, 2'5]$ $z_2 = [3,1]$.

Exercise 5: K-means segmentation and a linear SVM.

(a) A BI team clusters four business customers with $k = 2$:

$$A = (c, c), \quad B = (c, c+1),$$
$$C = (c+4, c+3), \quad D = (c+5, c+4).$$

The initial centroids are

$$m_1^{(0)} = (c-1, c), \qquad m_2^{(0)} = (c+5, c+3).$$

Compute the squared Euclidean distance of every point to both centroids and perform the first assignment step. Break ties in favor of the smaller cluster index. *(5 points)*

(b) Compute the updated centroids $m_1^{(1)}$ and $m_2^{(1)}$. *(4 points)*

(c) Perform one more assignment step, decide whether the algorithm has converged, and interpret the two clusters as business segments. *(3 points)*

Exercise. SS 26.

Matrikelnumber $\underset{a}{2}\ \underset{b}{2}\ \underset{c}{6}\ \underset{d}{3}\ \underset{e}{4}\ \underset{f}{7}$

$A=[6,6]$ $B=[6,7]$ $C=[10,9]$ $D=[11,10]$

$m_1^{(0)}=\{5,6\}$ $m_2^{(0)}=\{11,9\}$

a) $d_{Am_1^{(0)}}=\sqrt{(6-5)^2+(6-6)^2}=1$ $d_{Am_2^{(0)}}=\sqrt{(6-11)^2+(6-9)^2}=5'83$

$d_{Bm_1^{(0)}}=\sqrt{(6-5)^2+(7-6)^2}=1'41$ $d_{Bm_2^{(0)}}=\sqrt{(6-11)^2+(7-9)^2}=5'38$

$d_{Cm_1^{(0)}}=\sqrt{(10-5)^2+(9-6)^2}=5'83$ $d_{Cm_2^{(0)}}=\sqrt{(10-11)^2+(9-9)^2}=1$

$d_{Dm_1^{(0)}}=\sqrt{(11-5)^2+(10-6)^2}=7'2$ $d_{Dm_2^{(0)}}=\sqrt{(11-11)^2+(10-9)^2}=1$

$G_1=\{A,B\}$ $G_2=\{C,D\}$

b) $z_1=\left[\frac{6+6}{2},\frac{6+7}{2}\right]=[6,6'5]$ $z_2=\left[\frac{10+11}{2},\frac{9+10}{2}\right]=[10'5,9'5]$

c) $d_{AZ1}=\sqrt{(6-6)^2+(6'5-6)^2}=0'5$ $d_{AZ2}=\sqrt{(6-10'5)^2+(6-9'5)^2}$ >>

$d_{BZ1}=\sqrt{(6-6)^2+(7-6'5)^2}=0'5$ $d_{BZ2}=\sqrt{(6-10'5)^2+(7-9'5)^2}$ >>

$d_{CZ1}=\sqrt{(6-10)^2+(6'5-9)^2}$ >> $d_{CZ2}=\sqrt{(10-10'5)^2+(9-9'5)^2}$

$d_{DZ1}=\sqrt{(6-11)^2+(6'5-10)^2}$ >> $d_{DZ2}=\sqrt{(10'5-11)^2+(9'5-10)^2}$

$G_1=\{A,B\}$ $G_2=\{C,D\}$

$Z_1=[6,6'5]$ $Z_2=[10'5,9'5]$